Submission Summary

Conference Name
6th OPJU International Technology Conference (OTCON 6.0) on SMARTICON (Smart & Intelligent Computing for Next-Generation Technologies)

Track Name
Smart Computing for Artificial Intelligence, Machine Learning & Intelligent Systems

Paper ID
337

Paper Title
MamaMind: A Voice-Enabled, Family-Centred Mobile Application for Maternal Wellness, Paternal Support, and Infant Care Coordination Using Firebase and Generative AI

Abstract
Pregnancy and the postpartum period bring significant emotional, physical, and logistical challenges for new parents, particularly mothers. However, most existing digital applications focus on individual aspects of maternal wellness or infant care, offering limited support for family involvement. This paper presents MamaMind, a family-centred, cross-platform mobile application that integrates maternal wellness, paternal support, and infant care coordination into a unified system. The application features dedicated dashboards for mothers, fathers, and babies, along with a voice-enabled AI companion that allows mothers to interact through spoken commands and receive personalised responses. A Dad Coach provides fathers with practical suggestions to support mothers and participate in childcare. The application also incorporates shared family tasks, mood check-ins, feeding and sleep logs, diaper tracking, vaccination reminders, guided meditation, and self-care activities. Developed using Expo, React Native, and TypeScript, the system uses Firebase Authentication and Cloud Firestore for secure user management and real-time family data synchronisation. Generative AI capabilities are implemented through Google Gemini using an Express backend. MamaMind aims to reduce the burden of managing postpartum care by encouraging shared parental responsibility and providing accessible digital wellness support. The current implementation serves as a working proof-of-concept, with formal usability and performance evaluation planned as future work.

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Submission Files

MamaMind_Paper publication.pdf (376.4 Kb, 2026-11-01 21:24:14 GMT+5:30)